

FATIMA MAMU

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Boston, MA - Happy to relocate

PROFESSIONAL SUMMARY

AI Product Engineer focused on turning ambiguous problems into working systems, with experience building voice-based agents, real-time interaction systems, and structured knowledge workflows. Has shipped user-facing products used in clinical environments, combining product thinking with hands-on engineering to deliver practical, end-to-end solutions.

EDUCATION

Harvard University , Cambridge, USA — Master in Design Engineering	2023 – 2025
University College London , London, UK — Diploma in Computer Graphics, Vision and Imaging	2015 – 2016
Brunel University , London, UK — BEng, Electrical and Electronics Engineering	2010 – 2013

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C#, C++

AI & Machine Learning: LLM application development, retrieval-augmented generation (RAG), agentic workflows, prompt engineering, structured knowledge systems, evaluation and benchmarking

Real-Time & Voice Systems: streaming audio processing, speech-to-text pipelines, real-time interaction systems

Web & Application Development: Next.js, React, REST APIs, full-stack development

Data & Infrastructure: PostgreSQL, Supabase, pgvector, Google Cloud (Cloud Run, Cloud SQL), API integration

Prototyping & Creative Technology: Unity, TouchDesigner, Maya, AR/VR prototyping

WORK EXPERIENCE

Founding AI Engineer | The One Care, Boston, MA, USA | 2026 – Present

- Architected and deployed a production-ready voice-based caregiving AI agent, combining structured retrieval (pgvector/PostgreSQL) with a reasoning layer for multi-turn, context-aware interactions; evaluated on ~1,500 simulated queries, improving response relevance by ~25% over a baseline LLM.
- Built a real-time voice-to-structured-data pipeline converting unstructured caregiver speech into validated clinical records, with end-to-end latency optimized for live interaction workflows and schema-level validation to ensure consistency across extracted fields.
- Developed an automated ingestion pipeline using Gemini 2.5 Flash to classify caregiver experiences at scale, incorporating human-in-the-loop review to maintain high-quality knowledge indexing prior to retrieval.
- Deployed infrastructure on Google Cloud (Cloud Run, Cloud SQL, Secret Manager) and built a FastAPI-based evaluation harness with structured test cases covering ambiguous inputs, incomplete context, and conflicting caregiver reports.

AI Product Engineer | Lexi AI, Boston, MA, USA | 2025 – 2026

- Shipped customer-facing clinical interpretation features used across 4 facilities and 7,000+ conversations, improving clinician efficiency by 40% through real-time audio intelligence, visual cues, and workflow-integrated controls.
- Built and deployed a phone-based AI intake agent using Gemini Live to collect and structure patient data from low digital literacy users, with real-time speech processing and API-based routing to provider systems (JotForm) on Google Cloud.

- Conducted on-site clinician research and translated workflow insights into product and system improvements for real-world clinical environments.

Software Development Intern (Creative Technology) | Walt Disney Imagineering (R & D), Glendale, CA, USA | 2024

- Built internal tools including a synthetic dataset generation pipeline for 6-DoF pose estimation and an AR environment for pre-production staging, reducing setup time and improving spatial accuracy before physical installation.
- Developed real-time interactive experiences using Unity, Maya, and TouchDesigner, including a guest-driven water simulation for the *Moana: Journey of Water* exhibit at D23 that translated full-body motion tracking into responsive visual effects.
- Integrated robotics, real-time animation, and physical staging for a KUKA robot-based illusion, synchronizing TouchDesigner animations with LED outputs and presenting the final system to Bob Iger for project continuation approval.

Engineering Associate (Prototyper) | The Business People Limited | 2019 – 2023

- Led rapid prototyping of client-facing systems across agriculture, aviation, and healthcare, translating ambiguous operational challenges into functional prototypes used to validate product direction and reduce implementation risk prior to full-scale builds.
- Designed and built full-stack prototypes (Next.js, TypeScript, PostgreSQL), implementing end-to-end workflows including data modeling, API design, and user interaction flows to simulate real system behavior.
- Acted as the primary technical lead on client engagements, owning requirements discovery, system design, stakeholder communication, and coordination with external engineering teams for implementation.